

Five measurement errors that kill AI pilots

Why the pilot looked good, and the rollout did not.

Every one of these was found in our own work, in a single week, on a system we built ourselves. Four of the five were *our* errors, caught by our own checks before they cost anything. That is the point: these are not mistakes other people make.

The worked example is public and reproducible — every figure in this document can be recomputed from source at github.com/marcdebary/pythia.

A short paper on evaluating automation before you scale it.

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The problem with a good demo

A pilot is usually judged by watching it work. Someone runs ten examples, nine look right, and the decision is made. Then it goes into production and the benefit does not appear.

Nothing went wrong in the rollout. The measurement was wrong at the start. Ten examples cannot distinguish a system that is genuinely better from one that is merely agreeable, and "it looked right" is not a baseline.

We recently spent a week building a system to answer one narrow question: does a particular pricing signal beat a live market? The answer turned out to be no — three separate strategies, each disproved with a number. But the more useful output was the list of ways we nearly fooled ourselves along the way. Those five failure modes are not specific to pricing. They are how automation projects go wrong in general.

ERROR ONE

1 Counting the same thing twice

We had 245 measurements and treated them as 245 independent results. They were not. They were 46 events, each observed about five times. Repeated looks at the same event are not new evidence — but they make a statistic look five times more certain than it is.

WHAT IT DID TO THE NUMBER

Counting every row, the result was +\$0.66 per \$100 with a confidence interval of +0.46 to +0.86 — a clear win. Counting one result per event, as you would actually act on it, the same data gave +\$0.31 with an interval of -0.05 to +0.67. The interval now includes zero. The effect had not changed; only the honesty of the count had.

IN YOUR BUSINESS

A support assistant is evaluated on 500 tickets — but they come from 60 customers, and one customer's tickets all look alike. A document tool is tested on 200 pages from 12 contracts. A forecasting model is scored on 12 months of daily data that is really 12 independent months. In each case the sample is far smaller than it appears, and the confidence is borrowed.

The check: ask what the independent unit actually is — the customer, the contract, the month — and count those. If the number drops by a factor of five, your certainty should drop with it.

ERROR TWO

2 An assumption dressed as a result

Our model needed a value we did not have, so we estimated it and moved on. The system then reported an enormous advantage in one location. It was not an advantage; it was the estimate, reflected back at us as if it were a finding.

WHAT IT LOOKED LIKE

The system reported an 89 percentage point edge in San Francisco — an implausible figure that should have been read as an alarm, not an opportunity. The cause: two forecast sources disagreed by 11 °F, and our uncertainty estimate only knew about one of them. Once the disagreement between sources was included, the edge disappeared entirely.

IN YOUR BUSINESS

A savings case rests on "we assume 20 % time saved." A routing model rests on an assumed conversion rate. A staffing plan rests on an assumed error rate. The assumption enters at the top and comes out of the bottom as a business case — and by then nobody remembers it was a guess.

The check: every assumed number is labelled as assumed, and the result is recomputed at the plausible extremes. If the conclusion flips, you do not have a business case — you have an opinion with arithmetic attached.

ERROR THREE

3 Silent failure

The machine hosting our system restarted. The service came back and looked healthy. It wrote nothing for twenty-nine hours, and nobody noticed — because a log file goes quiet precisely when nothing is running. There was no error to see. Absence does not raise alarms.

WHAT IT COST

A day and a half of collection, which was exactly the data the scheduled evaluation needed. The status indicator was green throughout. It was reporting that the process was alive — not that it was doing anything.

IN YOUR BUSINESS

An integration stops receiving records after a credential expires. A nightly job runs and processes an empty file. A classifier quietly falls back to a default and labels everything the same way. Every dashboard stays green, because all of them monitor liveness rather than output.

The check: monitor the *age of the newest result*, not the health of the process. "The last record is four minutes old" is a fact. "Service: running" is not.

ERROR FOUR

4 Right, and still useless

Our weather model was well calibrated: when it said 27 %, the event happened about 23 % of the time. That is a genuinely good property, and we were pleased with it. It was also irrelevant, because the market's forecast was three times sharper on exactly the same events.

CALIBRATED IS NOT THE SAME AS VALUABLE

Our error score was 0.140. The benchmark we were trying to beat scored 0.041. Both were honest; one was far more decisive. Acting on ours would have lost \$81 per \$100 staked.

IN YOUR BUSINESS

A model that predicts churn accurately but only flags customers your account managers had already flagged. A document classifier that is 94 % accurate on a task where one rule of thumb gets 93 %. A forecast that is unbiased and too vague to schedule against. Accuracy against the truth is the wrong comparison. The right one is accuracy against *what you would have done anyway*.

The check: always score the new system and the current process on the same cases, side by side. Not against perfection — against the incumbent, including the boring incumbent of "an experienced person guessing."

ERROR FIVE

5 Value you cannot actually capture

This one is the most expensive, and the easiest to miss, because the analysis is correct right up to the end. We found a real, statistically defensible pricing advantage. We then checked whether we could ever have acted on it. We could not.

THE NUMBER THAT ENDED IT

Ahead of our order in the queue sat a median of 71,589 contracts; our order was 188. Of 46 attempts, 12 were filled — and the ones that filled had a *negative* edge (−0.090) while the ones that never filled had a positive one (+0.179). We were served only when the market had already moved against us.

IN YOUR BUSINESS

The tool saves four minutes per case, but the person still waits for the next case to arrive, so the hour is not recovered. The model identifies 300 upsell candidates and the team can call thirty. The automation clears the queue by noon and the downstream department still batches once a day. The benefit is real on the page and never lands in the accounts.

The check: before measuring the benefit, describe the mechanism by which it becomes money or time — who does what differently, and what constraint absorbs the slack. If nobody's day changes, nothing has been saved.

The order the questions must be asked in

Most evaluations start at step four. That is the mistake that produces confident wrong answers.

- 1 Is it calibrated?** When it says 70 %, does it happen roughly 70 % of the time? A system that cannot be trusted at face value cannot be reasoned about at all, and every number after this one is decoration.
- 2 Is it better than what you do now?** Scored on identical cases, against the actual incumbent. This is where most pilots quietly end — and finding that out in week two is worth far more than finding it out in year two.
- 3 Is the difference larger than the noise?** With the sample counted honestly. Small real differences need more evidence than anyone expects, and "it seems better" is not evidence.
- 4 Can you capture it?** Through a named mechanism, against a named constraint. This is the question that decides whether the project pays for itself.

None of this requires exotic tooling. It requires the discipline to run the steps in order, and the willingness to accept a "no" at step two.

Why our answer to "does it work?" is worth something

Anyone can show you a system that succeeded. The harder credential is a system that was measured honestly and reported a failure — including the measurer's own errors.

HYPOTHESIS TESTED	VERDICT	EVIDENCE
Buy when the reference price is higher	Rejected	−0.49 pp against a 4.00 pp cost hurdle
Post passively and collect the spread	Rejected	12 of 46 filled; the filled ones were the bad ones
Move to markets where execution is easy	Rejected	error 0.140 vs the benchmark's 0.041

Three plausible ideas, three weeks of runway saved, and no funded account lost proving it the expensive way. The code, the data model, the checks and the mistakes are all public.

What we do

DE BARY LLC builds and operates the execution layer for AI workflows — your workload, in your container, with your data staying where it is. We also do the unglamorous part that determines whether any of it was worth doing: measuring, before you scale, whether the thing actually beats what you already have.

If you have a pilot that looked good and has not yet paid for itself, that is exactly the conversation we are useful for.

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